

Problem

A mechanical system is modeled by a series RLC circuit. It is desired to produce an overdamped response with time constants 0.1 and 0.5 ms. If a series 50-k Ω resistor is used, find the values of L and C .

Step-by-step solution

Step 1 of 2

$$\begin{aligned}
 t_1 &= \frac{1}{|S_1|} = 0.1 \times 10^{-3} \\
 \text{Leads to } S_1 &= \frac{-1000}{0.1} = -10,000 \\
 t_2 &= \frac{1}{|S_2|} = 0.5 \times 10^{-3} \quad \text{Leads to } S_1 = -2000 \\
 S_1 &= -\alpha - \sqrt{\alpha^2 - \omega_0^2} \\
 S_2 &= -\alpha + \sqrt{\alpha^2 - \omega_0^2} \\
 S_1 + S_2 &= -2\alpha = -12,000 \\
 \text{Therefore } \alpha &= 6000 = \frac{R}{2L}
 \end{aligned}$$

Step 2 of 2

$$\begin{aligned}
 L &= \frac{R}{12,000} = \frac{50,000}{12,000} = 4.1674 \\
 S_2 &= -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -2,000 \\
 \alpha - \sqrt{\alpha^2 - \omega_0^2} &= 2,000 \\
 6000 - \sqrt{\alpha^2 - \omega_0^2} &= 2,000 \\
 \alpha^2 - \omega_0^2 &= 16 \times 10^6 \\
 \omega_0^2 &= \alpha^2 - 16 \times 10^6 = 36 \times 10^6 - 16 \times 10^6 \\
 \omega_0^2 &= 10^3 \sqrt{20} = \frac{1}{\sqrt{LC}} \\
 C &= \frac{1}{20 \times 10^6 \times 4.167} = 124 F
 \end{aligned}$$